

1. Explain Mean, Median and Mode in the context of customer income.

- Mean is the average income of all customers.
Median is the middle income when all incomes are arranged in order.
Mode is the income value that occurs most often.

2. Differentiate between Standard Deviation and Variance.

- Variance tells us how much the values are spread from the average.
Standard Deviation also tells us the spread of the data.
Standard deviation is the square root of variance.

3. What is a Random Variable? Give one example from the dataset.

- A random variable is a value that can change from one customer to another.
It can be a number or a category.
For example, Loan_Amount is a random variable in our dataset.
Its value can be different for every customer.

4. Explain Conditional Probability in terms of loan defaults.

- Conditional probability means finding the chance of an event when we already know another condition.
For example, we can find the chance that a customer will default when their credit score is below 600.

5. Define Bayes Theorem and mention how banks can apply it.

- Bayes Theorem is used to find the probability of an event using some known information.
Banks can use it to estimate the chance that a customer may default on a loan.
They can use information like income, credit score and loan amount.

6. Differentiate between Empirical Probability and Theoretical Probability.

- Theoretical probability is based on mathematical calculation.
Empirical probability is calculated using actual data or experiments.
For example, the probability of getting heads from a coin is theoretical.
The percentage of loan defaults in our dataset is empirical probability.

7. What is a Poisson Distribution? Give a business example.

- Poisson Distribution is used to find how many times an event happens in a fixed time or place.

For example, a bank can use it to study the number of loan applications received in one hour.

It is useful when events happen randomly.

8. Write a short note on Eigenvalues and Eigenvectors in data analysis.

- Eigenvalues and eigenvectors are concepts used in linear algebra and data analysis.

They help us understand important directions and patterns in data.

They are commonly used in PCA (Principal Component Analysis).